$$_1H^1 + {}_0n^1 \rightarrow iD^z + h\nu$$

Now the mass of a deuterium atom is 2.0147 a.m.u., whereas the sum of the masses of the hydrogen atom and the neutron amounts to 1.00813 + 1.00895 = 2.0171 a.m.u. The mass of the deuterium atom is actually smaller than the sum of the masses of its component parts, and this difference is the mass defect, which amounts to 0.0024 a.m.u. (2.4 a.m.u.$^{-3}$). But expressed in terms of its energy equivalent, this mass defect represents a binding energy of about 2.2 Mev, which is exactly the energy released in the form of a photon, as we have already pointed out. Thus we see that there are two totally independent methods for the calculation of the binding energy, and both methods yield the same answer, and therefore provide the best possible proof of the truth of the principle of the equivalence of mass and energy. This is an especially important fact, for here we are not dealing with electric fields, with reference to which this principle was well known even in pre-relativistic times, but with fields of a totally different nature.

The findings just discussed show also why the atomic weights of the elements are not integral multiples of a basic unit, as Prout had conjectured. In the first place, the mass of the proton and the mass of the neutron differ slightly from each other. Furthermore, when they combine, a fraction of the sum total of their masses disappears; this fraction corresponds to their binding energy. So even if the mass of the proton were exactly equal to the mass of the neutron, the atomic weights of the elements would still not be integral multiples of a basic unit. This explains the slight deviations from integral numbers in the atomic weights of the lighter elements. But the very considerable deviations of the atomic weights of the heavier elements must be explained in another way. These deviations are more apparent than real and are due to the fact that the natural elements are, mostly, mixtures of various isotopes. Each isotope consists of atoms with a nucleus of a specific type, the mass number of which is almost exactly an integer. But the average mass number of this mixture of isotopes is not an integer.

Thus we have established the desired criterion for the stability of a nucleus. A nucleus holds together because an amount of work would be required in order to break it up into its component parts. The work required is a measure of the binding energy, which, in turn, is equal to the mass defect of the nucleus, expressed in terms of energy. A nucleus would not be stable if it were possible to break it up without doing any work. We must, however, add that the other conservation laws must be taken into consideration too. According to the law of the conservation of electric charge, an atom cannot undergo any